GRAVITON (CGC University Mohali)

KneeGait-Edge

Democratizing Gait Intelligence

A ₹516 edge-AI wearable for accessible gait analysis



Team Astronauts

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THE ACCESS GAP

Gait analysis is locked behind labs

Traditional approach

- Specialized equipment
- Expensive infrastructure
- Laboratory-dependent
- Limited access

650M+ adults live with knee osteoarthritis worldwide

$20K–$50K optical motion-capture systems (Vicon, Qualisys)

₹516 our single-IMU edge-AI prototype

Can gait intelligence become portable, accessible and affordable?

The need: Portable • Affordable • Real-time • Accessible

THE PROTOTYPE

Meet KneeGait-Edge

A wearable edge-AI prototype that tracks shank motion during walking with a single IMU strapped to the shin.

Mounted on the flat anteromedial tibia to limit soft-tissue wobble.

From body movement to real-time gait intelligence.

01 SENSE
Shank IMU captures movement

02 COMPUTE
ESP32 processes in real time

03 ANALYZE
Gait features are extracted

04 VISUALIZE
BLE streams to the browser

SYSTEM FLOW

From Motion to Gait Intelligence

SHANK IMU — ESP32 — SENSOR FUSION — STRIDE FEATURES — RANDOM FOREST — BLE DASHBOARD

RESEARCH BACKED

Bytyqi et al. (2014)

Clinical thresholds

Peak swing flexion ≈ 41.9° in medial knee OA vs ≈ 58° in healthy controls; basis for the < 45° screening flag.

Rhudy et al. (2024)

Sensor fusion

Magnetometer-free complementary filter (α = 0.96 at 50 Hz); reported joint-angle RMSE ≤ 3.2° vs motion capture.

Song & Baxter (2025)

Placement & validation

Tibia-mounted IMUs limit soft-tissue artifact; reported Rxy ≥ 0.90 against optical motion capture.

Literature figures come from knee-angle studies, not from this prototype's own measurements.

HARDWARE

Simple Hardware. Smart Processing.

ESP32 NodeMCU

Processing + BLE

1 × MPU-6050

6-axis motion sensing on the shank

USB power bank

Portable power

Elastic bandage

Wearable mounting

I²C BUS

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SDA → GPIO 21    SCL → GPIO 22    Shank IMU → 0x68 (AD0 low) / 0x69 (AD0 high)
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Cost democratization: capable sensing from components anyone can buy.

₹516 prototype cost

SIGNAL FUSION

Turning Raw Motion Into Shank Angle

1

ACCELEROMETER TILT

Inclination from gravity: atan2(−ax, √(ay²+az²))

2

GYROSCOPE INTEGRATION

Rotation over time: θ(t−Δt) + ω·Δt

3

COMPLEMENTARY FILTER

Fused angle with less noise and drift

$$\theta(t) = 0.96 \times [\theta(t-\Delta t) + \omega \cdot \Delta t] + 0.04 \times \theta_{acc}(t)$$

50 Hz sampling • Δt = 20 ms • α = 0.96

One IMU gives shank inclination θshank, a proxy for knee motion. True knee flexion, |θthigh − θshank|, needs a second (thigh) sensor, which is planned next.

From Strides to Features

HEEL-STRIKE EVENT DETECTION

≥ 1.45 g
Shank impact spike marks a heel strike

0.40 s
Refractory window rejects bounce and chatter

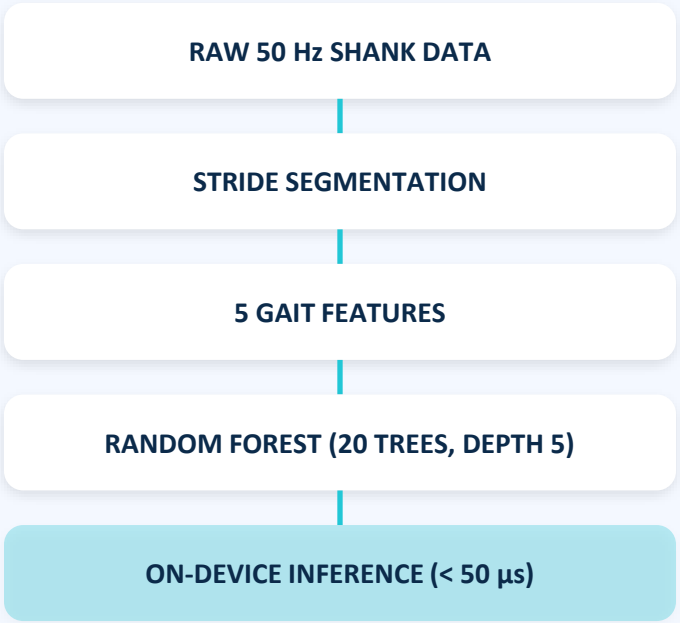
0.55–2.40 s
Valid stride duration (physiological bounds)

FIVE GAIT FEATURES PER STRIDE

1. **Peak swing angle**
   Max shank angle during the swing phase
2. **Sagittal range of motion**
   Max − min angle across the stride
3. **Stance / swing ratio**
   Weight-bearing time ÷ swing time
4. **Peak angular velocity**
   Max rate of rotation, max $|d\theta/dt|$
5. **Cadence**
   Steps per minute = 120 ÷ stride time (s)

EDGE AI

Intelligence at the Edge



3-CLASS RISK OUTPUT

Normal gait
Peak swing flexion ≥ 49.4° · continue monitoring

Mild functional restriction
45.0° ≤ peak swing flexion < 49.4° · follow up

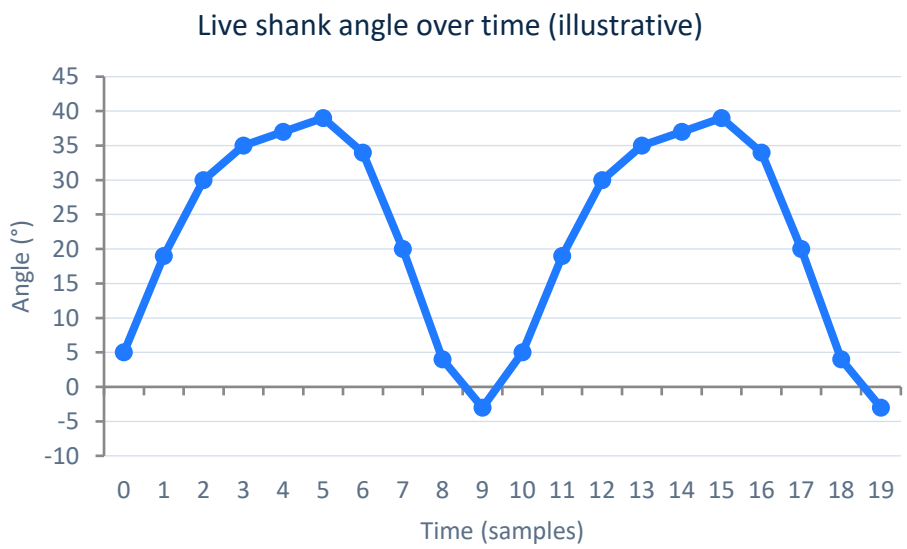
Severe antalgic stiff gait
Peak swing flexion < 45.0° · physician review

Random Forest → portable C header → ESP32 inference. No cloud needed for core inference.

Trained on 20,000 synthetic strides; thresholds are knee-flexion references. Validation on real gait data is a next step.

LIVE EXPERIENCE

See the Gait in Real Time



Illustrative waveform; not patient data.

Web Bluetooth
Runs in Chrome or Edge, no app install

Risk badge
Green / amber / red triage at a glance

Reports
Session analytics, CSV export, printable PDF

BIOMECHANICAL CONTEXT

From Movement to Meaning

≈ 58°

Healthy mean peak swing flexion (Bytyqi 2014)

41.9°

Mean in moderate-to-severe medial knee OA

< 45°

Screening flag for stiff-gait risk, not a diagnosis

GAIT CYCLE

1 STANCE — 2 TOE-OFF — 3 MID-SWING — 4 HEEL STRIKE

Research / hackathon prototype, not intended for medical diagnosis. Shank-only angles are a proxy for knee flexion, so thresholds need recalibration on real data.

ROADMAP

From Prototype to Scalable Gait AI

NOW

- ₹516 prototype
- Single shank IMU
- ESP32 edge AI
- BLE web dashboard

NEXT

- Add thigh IMU for true knee flexion
- Real gait data and model validation
- Compare with optical capture
- Improved enclosure

FUTURE

- Clinical validation
- Bilateral left/right nodes
- Insole force sensors
- Cloud / EHR telehealth

POTENTIAL APPLICATIONS

Research • Physiotherapy support • Gait screening • Remote monitoring • Resource-constrained settings

CREDITS

References & Acknowledgements

RESEARCH REFERENCES

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Note: this is a hackathon/research prototype. Dashboard values shown are illustrative, not patient results.

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